

Jonathan chang

jonathan.archiechang@gmail.com | 0420 214 456 | [LinkedIn](#) | [Portfolio](#) | Sydney, NSW

WORK EXPERIENCE

Optik Consultancy

Nov. 2025 – Present

Supervisor & Project Management Lead

Botany, NSW

- Promoted to supervise incoming internship teams, overseeing project delivery and mentoring junior engineers.
- Led a cross-disciplinary team to develop a hospital technology prototype, collaborating with stakeholders at Woolcock Institute, integrating Arduino sensors with a React/Electron desktop app.
- Architected the electronics subsystem and guided interface and software development.

The Movement Church

Nov. 2024 – Present

Connect group leader & Technical volunteer,

Sydney, NSW

- Developed backend features for a newcomer onboarding app using Node.js, TypeScript, and REST APIs.
- Led a connect group of 25 members and onboarded 50+ newcomers.

Coles

Mar. 2021 – Present

Grocery Department 2IC & Duty Manager

Engadine, NSW

- Managed department operations and supported store-wide decision-making as acting Duty Manager.
- Contributed to the grocery department becoming the top-performing in the region (23% growth).

PROJECTS

F1 Monaco GP 2024 Live Race Replay Dashboard | Personal Project | Completed

- Built an interactive real-time race replay using Python, Dash, and Plotly, processing 673,000+ GPS telemetry data points across all 20 drivers on the Monaco circuit.
- Engineered a data pipeline with chunked loading and cubic spline interpolation for smooth 10Hz car movement, optimised for memory efficiency at scale.
- Containerised with Docker for one-command deployment; includes live pit stop alerts, safety car/flag status, and official 2024 team livery colours.

CLARA: Conversational Learning Agent for Requirements Analysis | UTS Honours Thesis | In Progress

- Building an AI-powered requirements elicitation tool using Python and the Gemini API.
- Engineered a multi-turn dialogue system with stateful conversation history, exploring context stuffing and knowledge graph approaches for information retrieval.
- Multi-model fallback architecture for reliability in production-like conditions.

EDUCATION

University of Technology Sydney

Dec, 2026

Bachelor of Engineering (Honours)

Sydney, NSW

- **WAM:** 73.48 | **GPA:** 5.29
- **Academic achievements:** HD in Software Architecture (92), HD in Embedded Systems (92), HD in Software Analysis Studio (91)

SKILLS & INTERESTS

- **Technologies:** Node.js, SQL, Java, C, C++, Python, Rest API, HTML, CSS, Typescript, Javascript, NoSQL, React, Electron
- **Skills:** Web Development, Data Engineering, Embedded Systems, Software Architecture, Team Leadership, Stakeholder Communication.
- **Interests:** Weightlifting, Motorsports, Volleyball, Video Games, Music instruments